■ Vellum Web: Fernly Botanical Edition

Dedicated Standalone Academic Research Studio — User & Technical Requirements Manual

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Release: Latest Standalone Web App (v3.0)	**Host & Port:** http://localhost:3002
Tech Stack: Next.js 16 (Turbopack) / React 19	**License:** 100% Free Open-Access (MIT)

1. System Overview

This distribution contains **only the latest standalone web application (Vellum Fernly Botanical Edition)**. It combines high-performance interactive bibliometrics with privacy-first evidence synthesis into a single browser-based workstation running on port 3002.

2. System & Hardware Requirements

Category	Requirement	Details & Guidance
Operating System	Windows 10 / 11 (64-bit)	Compatible with Windows 10 (1903+) and Windows 11
Runtime	Node.js 18.0.0 or higher	Download LTS version (v20+) from https://nodejs.org with PATH checked
RAM	4 GB min / 8 GB rec	8 GB optimal when screening datasets exceeding 3,000 candidate records
Web Browser	Chromium / Firefox / Edge	Ensure hardware acceleration is enabled for smooth force-graph canvas physics
Network	Outbound HTTPS (443)	Connects to OpenAlex, arXiv, Semantic Scholar, and Crossref APIs

3. 1-Click Launch Instructions

To start the application, double-click the **START_VELLUM_WEB.bat** launcher in the root of the folder. The script will:

1. Verify the Node.js runtime environment.
2. Launch the Next.js 16 Turbopack development server on **http://localhost:3002**.
3. Automatically open your default web browser directly to the research studio.

To terminate the session, simply close the command prompt window.

4. Core Features & Scientific Modules

■■ **Connected Papers Force-Directed Citation Network:** Visualizes research clusters using HTML5 canvas with real-time Coulomb-Hooke physics. The origin paper is highlighted by a glowing purple ring, node sizes scale with lifetime citations, and node colors follow a continuous mint-to-pine publication year gradient. Features direct tabs for **Prior Works** (foundational ancestry) and **Derivative Works** (subsequent breakthroughs), plus a 1-click *'Build graph from this paper'* button to branch the network from any node.

■ **Desktop-Style Clause-by-Clause Boolean Search:** Located under Academic Search. Allows researchers to build exact queries with an initial WHERE clause followed by AND, OR, and NOT connectors. Field dropdowns automatically map terms to target index tokens ([Title/Abstract], au:, year:, topic:) with live compilation and 1-click execution.

■ **ASReview Active-Learning Screening Hub:** Machine-learning assisted title and abstract screening. Upload candidate bibliographies (.csv, .ris, .bib) with automatic multi-source deduplication. Employs TF-IDF feature extraction with Naive Bayes probability ranking, keyboard hotkeys ([I] Include, [E] Exclude, [U] Uncertain), session saving/switching, and live recall stopping curves.

■ **Multi-Tier Batch PDF Downloader:** Integrated batch downloader that queries Unpaywall, Open Access repositories, and Semantic Scholar with automated failover and paywall safeguards. Streams PDFs directly into client memory and browser downloads.

■ **Grounded Tri-Pane Evidence Synthesis Notebooks:** Local-first systematic review workspaces. 100% private and persistent using browser IndexedDB (Dexie.js). Left pane stores sources and PDFs; center pane provides an interactive document reader; right pane serves as an evidence synthesis studio for collecting verbatim claims and generating citation exports.

5. Troubleshooting & Technical Guidance

- **Port Occupied (EADDRINUSE):** If port 3002 is already occupied by a previous session, run Get-Process -Name node | Stop-Process -Force in PowerShell, then re-launch START_VELLUM_WEB.bat.
- **Data Privacy & Storage:** All notebooks, citations, and screening sessions are stored locally in your browser's IndexedDB. No accounts or cloud logins are required.
- **Workspace Reset:** Click **'Clean Workspace'** on the Dashboard to safely wipe or reset local database caches.